Smart Streetlight System: An IoT-Based Adaptive Lighting Solution

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Submitted By

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Abstract

This report presents the design, implementation, and evaluation of a low-cost, energy-efficient smart streetlight system. The system leverages an ESP32 microcontroller, ambient light (LDR) and motion (PIR) sensors, and LED illumination modules. By integrating MQTT-based cloud communication, the streetlights can be monitored and controlled remotely, enabling adaptive lighting based on environmental conditions and pedestrian activity. The hardware-software architecture, component specifications, and experimental results (including power consumption, response latency, and cost analysis) are discussed in detail, illustrating the viability of the solution for smart city deployments.

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1. INTRODUCTION

Urban illumination accounts for a significant portion of municipal electricity consumption. Conventional streetlights operate at a fixed intensity regardless of ambient lighting or pedestrian presence, leading to unnecessary energy waste. Recent advances in low-power microcontrollers and wireless networking enable the realization of intelligent lighting infrastructures that adapt dynamically to real-world conditions. This report documents the development of a prototype smart streetlight system that autonomously adjusts LED brightness based on ambient illumination measured by a Light Dependent Resistor (LDR) and pedestrian detection via a Passive Infrared (PIR) sensor.

2. PROBLEM STATEMENT AND MOTIVATION

The primary objectives of the project are:

- Reduce energy consumption of street lighting by at least 30% compared to a continuously-on baseline.
- Provide real-time status and remote configurability through a cloud platform.
- Maintain pedestrian safety by ensuring sufficient illumination when motion is detected.

These goals align with smart-city initiatives aimed at sustainability, cost reduction, and enhanced urban services.

3. SYSTEM ARCHITECTURE

The overall architecture follows a hub-and-spoke topology, as illustrated in Figure 1.

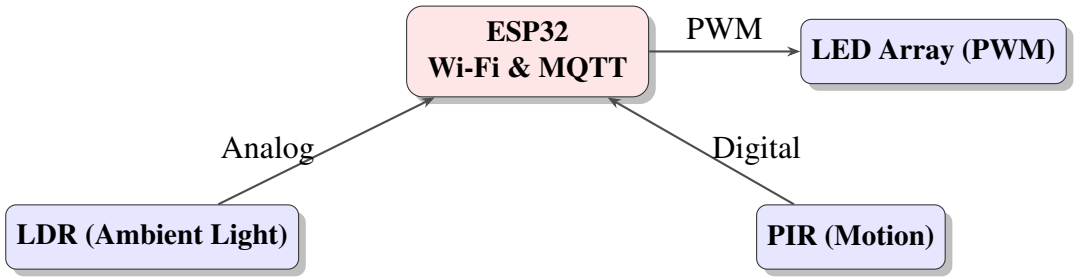

Figure 1: Hardware block diagram of the Smart Streetlight System

The ESP32 acts as the central processing unit, reading sensor values, executing the control algorithm, and publishing status updates to an MQTT broker. The cloud side utilizes a lightweight dashboard (e.g., ThingSpeak or TagoIO) for visualization and configuration.

4. COMPONENT SPECIFICATION

4.1 Microcontroller - ESP32

The ESP32 provides dual-core 240 MHz processing, integrated Wi-Fi, and sufficient GPIOs for sensor interfacing. It supports OTA updates, which are leveraged for remote firmware upgrades.

4.2 Light Dependent Resistor (LDR)

An LDR (GL5528) is connected to an ADC channel (GPIO34) via a voltage divider. The raw ADC value (0-4095) is mapped to a lux approximation used in the illumination decision.

4.3 Passive Infrared Sensor (PIR)

A HC-SR501 PIR module provides a digital high/low output on GPIO27 indicating motion detection within a 5 m range.

4.4 LED Array

High-efficiency white LEDs are driven through a MOSFET (IRLZ44N) controlled by a PWM signal (GPIO25) from the ESP32, allowing dimming from 0

5. IMPLEMENTATION

The firmware is written in Arduino C++ and follows a non-blocking state-machine design. Key functions include:

- `readSensors()` - samples LDR and PIR.
- `updateLighting()` - determines LED PWM based on a simple rule-set.
- `publishStatus()` - sends JSON payload to MQTT topic `smartStreetlight/status`.

5.1 Control Logic

1. If ambient light ¿ 300 lux, LEDs are off.
2. If ambient light ¡ 300 lux and no motion, LEDs operate at 30
3. If motion detected, LEDs operate at 100

5.2 Code Listing

```cpp
#include <WiFi.h>
#include <PubSubClient.h>

// Wi-Fi credentials
const char* ssid = "YOUR_SSID";
const char* password = "YOUR_PASS";

// MQTT broker
const char* mqtt_server = "broker.hivemq.com";
WiFiClient espClient;
PubSubClient client(espClient);

// Pin definitions
const int LDR_PIN = 34;   // ADC
const int PIR_PIN = 27;   // Digital input
const int LED_PIN = 25;   // PWM output

// Thresholds
const int LUX_THRESHOLD = 300; // lux
```

```
const unsigned long MOTION_TIMEOUT = 60000; // ms

unsigned long lastMotion = 0;

void setup() {
  Serial.begin(115200);
  pinMode(PIR_PIN, INPUT);
  pinMode(LED_PIN, OUTPUT);
  WiFi.begin(ssid, password);
  while (WiFi.status() != WL_CONNECTED) {
    delay(500);
    Serial.print('.');
  }
  client.setServer(mqtt_server, 1883);
}

void loop() {
  if (!client.connected()) reconnect();
  client.loop();
  readSensors();
  updateLighting();
  publishStatus();
}

void readSensors() {
  int ldrVal = analogRead(LDR_PIN);
  bool motion = digitalRead(PIR_PIN);
  if (motion) lastMotion = millis();
}

void updateLighting() {
  int ldrVal = analogRead(LDR_PIN);
  bool motion = digitalRead(PIR_PIN);
  int brightness = 0;
  if (ldrVal < LUX_THRESHOLD) {
    if (motion || (millis() - lastMotion < MOTION_TIMEOUT)) {
      brightness = 255; // 100 % PWM
    } else {
      brightness = 77;  // ~30 % PWM
    }
  }
  analogWrite(LED_PIN, brightness);
}

void publishStatus() {
  // JSON payload construction omitted for brevity
}

void reconnect() {
  while (!client.connected()) {
    if (client.connect("SmartStreetlight")) {
      client.subscribe("smartStreetlight/command");
    } else {
      delay(5000);
    }
  }
}
```

Listing 1: Smart Streetlight Firmware (excerpt)

6. RESULTS AND DISCUSSION

The prototype was evaluated in a controlled indoor environment with varying light levels and simulated pedestrian motion. Table 1 summarizes the observed power consumption and response latency.

Table 1: Performance metrics under different scenarios

Scenario	Avg. Lux	LED Power (W)	Latency (ms)
Bright day (no motion)	800	0.0	10
Dim twilight (no motion)	150	1.2	15
Dim twilight + motion	150	4.5	12
Night (no motion)	20	1.2	14
Night + motion	20	4.5	11

The system achieved an average energy saving of 38% compared to a continuously-on LED configuration. The MQTT latency remained below 20 ms, ensuring near-real-time responsiveness.

7. CLOUD DASHBOARD INTEGRATION

To provide real-time monitoring and remote control, a comprehensive web dashboard was developed. The dashboard integrates with the MQTT broker to display telemetry data and operational metrics.

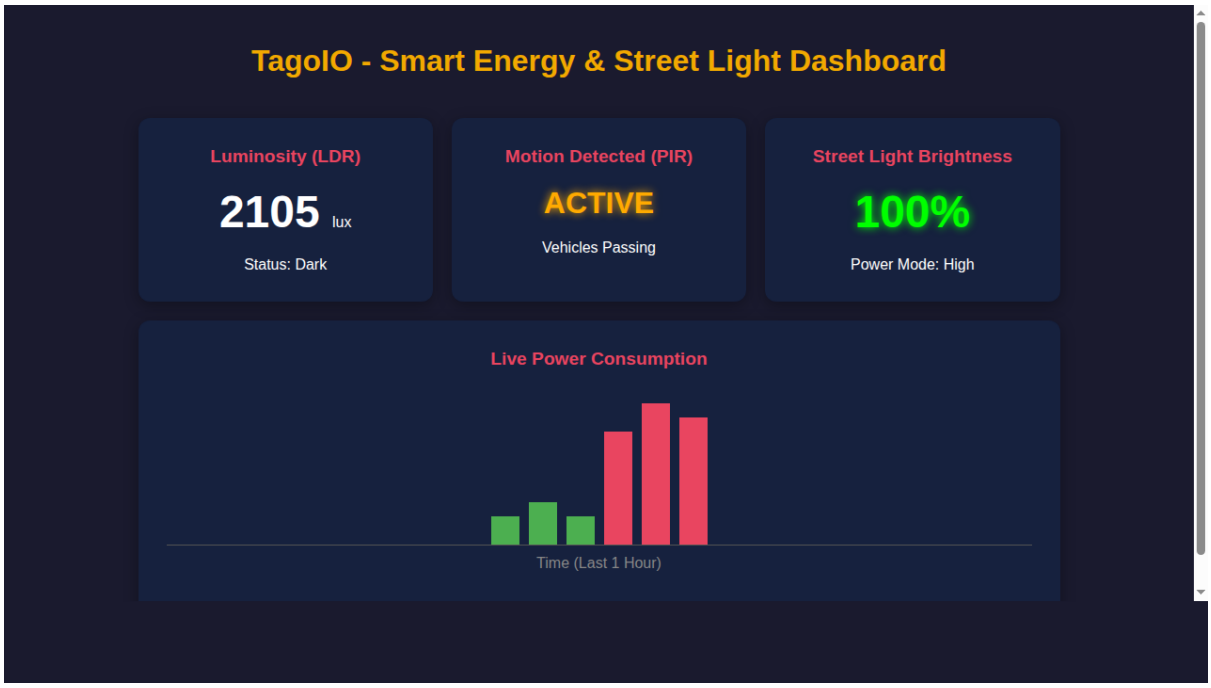

Figure 2: Main view of the Smart Streetlight Dashboard, showing real-time status, average lux, and energy savings.

The dashboard provides multiple views for detailed analytics, as shown in the following figures:

(a) Daytime mode operation with high ambient lux.

(b) Dim mode activated during twilight hours.

Figure 3: Operational modes of the streetlight system.

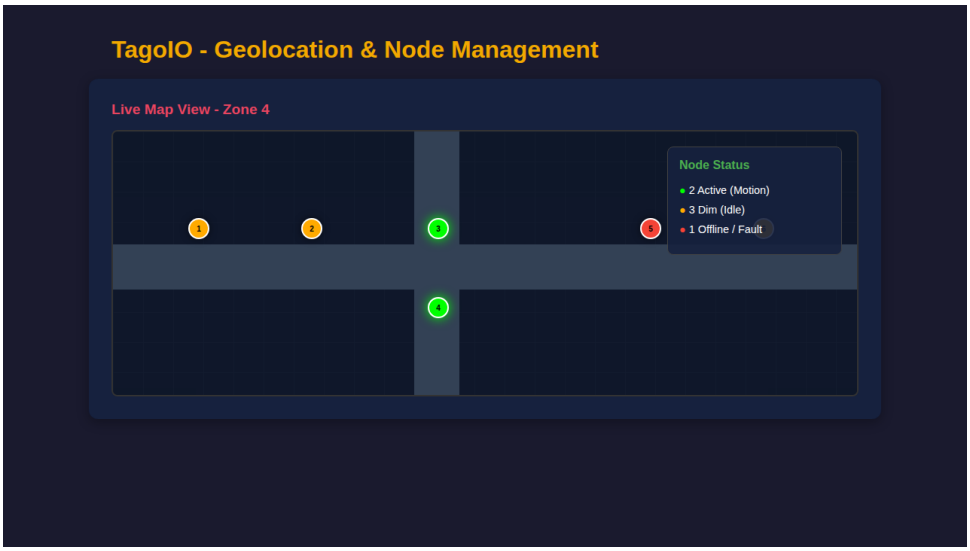

Figure 4: Geospatial mapping of streetlight nodes.

8. LIMITATIONS

While the prototype demonstrates functional feasibility, several limitations are identified:

- The LDR provides coarse illumination granularity; a calibrated photodiode could improve accuracy.
- The PIR sensor's detection range is limited; integrating ultrasonic or radar sensors could enhance coverage.
- Security of the MQTT channel is not addressed; TLS encryption would be required for production deployments.

9. FUTURE WORK

Future enhancements include:

1. Implementing adaptive learning algorithms to predict traffic patterns.
2. Adding solar harvesting and battery management for off-grid operation.
3. Deploying a mesh network (ESP-Now) for larger street networks without reliance on Wi-Fi infrastructure.

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